

Worksheet 5D: Trigonometric Identities and Equations

AIML Foundations Mathematics

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***What This Worksheet Is About** Trigonometric identities are equations that are true for all values of the variable. They're not problems to solve — they're tools that let us simplify expressions, prove relationships, and solve equations. These identities appear constantly in physics, engineering, and signal processing. Mastering them gives you powerful algebraic tools.*

Part A: Fundamental Identities (18 problems)

The Pythagorean Identity

From the unit circle, we know:

$$\sin^2 \theta + \cos^2 \theta = 1$$

Derived forms: $-\sin^2 \theta = 1 - \cos^2 \theta$ $-\cos^2 \theta = 1 - \sin^2 \theta$

Simplify using the Pythagorean identity:

1. $\sin^2(x) + \cos^2(x)$
2. $1 - \sin^2(x)$
3. $1 - \cos^2(x)$
4. $\frac{\sin^2(x)}{1 - \cos^2(x)}$
5. $\frac{1 - \sin^2(x)}{\cos(x)}$
6. $(\sin(x) + \cos(x))^2$ *Hint: Expand, then use the identity.*

Quotient and Reciprocal Identities

$$\begin{aligned} \tan \theta &= \frac{\sin \theta}{\cos \theta} & \cot \theta &= \frac{\cos \theta}{\sin \theta} = \frac{1}{\tan \theta} \\ \sec \theta &= \frac{1}{\cos \theta} & \csc \theta &= \frac{1}{\sin \theta} \end{aligned}$$

Simplify:

7. $\frac{\sin(x)}{\cos(x)}$
8. $\sin(x) \cdot \sec(x)$

9. $\cos(x) \cdot \csc(x)$
10. $\tan(x) \cdot \cos(x)$
11. $\frac{\sec(x)}{\csc(x)}$
12. $\sin^2(x) + \sin^2(x) \tan^2(x)$ *Hint: Factor out $\sin^2(x)$.*

More Pythagorean Identities

Dividing $\sin^2 \theta + \cos^2 \theta = 1$ by $\cos^2 \theta$:

$$\tan^2 \theta + 1 = \sec^2 \theta$$

Dividing by $\sin^2 \theta$:

$$1 + \cot^2 \theta = \csc^2 \theta$$

Simplify:

13. $\sec^2(x) - 1$
14. $\csc^2(x) - 1$
15. $\sec^2(x) - \tan^2(x)$
16. $\frac{\tan^2(x)}{\sec^2(x)-1}$
17. $\tan^2(x) + \cot^2(x) + 2$ *Hint: Rewrite using $\sec^2 - 1$ and $\csc^2 - 1$.*
18. $\frac{1-\cos^2(x)}{\sin(x)}$

Part B: Sum and Difference Formulas (20 problems)

The Formulas

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

Memory aid: - Sine: "sine-cosine, cosine-sine" with SAME sign as the original - Cosine: "cosine-cosine, sine-sine" with OPPOSITE sign

Find exact values using sum/difference formulas:

19. $\sin(75^\circ) = \sin(45^\circ + 30^\circ)$
20. $\cos(75^\circ)$

21. $\sin(15^\circ) = \sin(45^\circ - 30^\circ)$
22. $\cos(15^\circ)$
23. $\tan(75^\circ)$
24. $\sin(105^\circ)$ — Write as $60^\circ + 45^\circ$
25. $\cos(105^\circ)$ **Verify identities:**
26. Show that $\sin(x + \frac{\pi}{2}) = \cos(x)$
27. Show that $\cos(x - \frac{\pi}{2}) = \sin(x)$
28. Show that $\sin(\pi - x) = \sin(x)$
29. Show that $\cos(\pi - x) = -\cos(x)$ **Science Connection: Phase in Wave Addition**
30. When two waves of the same frequency but different phases combine:

$$y_1 = A \sin(\omega t) \quad \text{and} \quad y_2 = A \sin(\omega t + \phi)$$

Using the sum formula on y_2 : $y_2 = A[\sin(\omega t) \cos \phi + \cos(\omega t) \sin \phi]$ $y_1 + y_2 = A \sin(\omega t) + A \sin(\omega t) \cos \phi + A \cos(\omega t) \sin \phi = A(1 + \cos \phi) \sin(\omega t) + A \sin \phi \cos(\omega t)$ If $\phi = \frac{\pi}{2}$, simplify this expression.

31. If $\phi = \pi$ (waves completely out of phase), what is $y_1 + y_2$? **Science Connection: Rotation Matrices**
32. In computer graphics and robotics, rotation by angle θ uses the matrix:

$$R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

To rotate first by α , then by β , you multiply matrices. The result should be rotation by $\alpha + \beta$. This is equivalent to showing: $\cos(\alpha) \cos(\beta) - \sin(\alpha) \sin(\beta) = \cos(\alpha + \beta)$ Verify this matches the sum formula for cosine!

33. Rotating a point $(1, 0)$ by 30° , then by 60° , should give the same result as rotating by 90° .
 - (a) What is $\cos(90^\circ)$ and $\sin(90^\circ)$?
 - (b) Verify that $\cos(30^\circ) \cos(60^\circ) - \sin(30^\circ) \sin(60^\circ) = \cos(90^\circ)$

Astronomy: Orbital Mechanics

34. The position of a planet in an elliptical orbit involves angles that add and subtract. If a satellite's angular position is the sum of two angles: $\theta_{total} = \theta_1 + \theta_2$ where $\theta_1 = 45^\circ$ and $\theta_2 = 60^\circ$, find $\cos(\theta_{total})$ and $\sin(\theta_{total})$ exactly.

Part C: Double Angle and Half Angle Formulas (18 problems)

Double Angle Formulas

Setting $A = B$ in the sum formulas:

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$$

$$\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Simplify:

35. $2 \sin(30^\circ) \cos(30^\circ)$

36. $\cos^2(45^\circ) - \sin^2(45^\circ)$

37. $2 \cos^2(30^\circ) - 1$

38. $1 - 2 \sin^2(60^\circ)$ **Find exact values:**

39. If $\sin \theta = \frac{3}{5}$ and θ is in Quadrant I, find:

(a) $\cos \theta$

(b) $\sin(2\theta)$

(c) $\cos(2\theta)$

40. If $\cos \theta = \frac{5}{13}$ and θ is in Quadrant I, find $\sin(2\theta)$ and $\cos(2\theta)$.

Half Angle Formulas

$$\sin\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 - \cos \theta}{2}}$$

$$\cos\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

(Sign depends on which quadrant $\frac{\theta}{2}$ is in) **Find exact values:**

41. $\sin(22.5^\circ)$ — *Use $\theta = 45^\circ$ *

42. $\cos(22.5^\circ)$

43. $\sin(15^\circ)$ — *Use $\theta = 30^\circ$ * — Compare to your answer from Problem 21!

44. $\cos(15^\circ)$ **Science Connection: Power Reduction** In electronics, it's often necessary to convert between powers of trig functions and multiple angles.

45. The power reduction formulas come from solving the double-angle formulas:

$$\sin^2 \theta = \frac{1 - \cos(2\theta)}{2} \quad \cos^2 \theta = \frac{1 + \cos(2\theta)}{2}$$

Verify by expanding $\cos(2\theta) = 1 - 2 \sin^2 \theta$ and solving for $\sin^2 \theta$.

46. In AC circuits, power is proportional to $\sin^2(\omega t)$ or $\cos^2(\omega t)$. Using the power reduction formula, show that: $\sin^2(\omega t) = \frac{1}{2} - \frac{1}{2} \cos(2\omega t)$ What does this tell you about the frequency of power oscillation compared to voltage oscillation? **Science Connection: Interference Patterns**

47. When two identical beams of light interfere, the intensity is proportional to:

$$I \propto \cos^2\left(\frac{\phi}{2}\right)$$

where ϕ is the phase difference. Using the half-angle relationship, rewrite this as:
 $I \propto \frac{1 + \cos \phi}{2}$

48. In Young's double-slit experiment, the phase difference is $\phi = \frac{2\pi d \sin \theta}{\lambda}$ where d is slit separation, θ is angle from center, and λ is wavelength. At what values of ϕ do you get: - (a) Maximum intensity (bright fringes)? - (b) Minimum intensity (dark fringes)? **Science Connection: Lithography Interference**

49. Interference lithography creates patterns by interfering two laser beams. The pattern has period:

$$p = \frac{\lambda}{2 \sin \theta}$$

where θ is the half-angle between the beams. Using $\lambda = 193$ nm and $\theta = 30^\circ$, what pattern period can be achieved?

50. To create 10 nm features, you need $p = 20$ nm. If $\lambda = 13.5$ nm (EUV), what angle θ is needed?

Part D: Solving Trigonometric Equations (20 problems)

Strategy

1. Isolate the trig function if possible
2. Use identities to simplify
3. Find all solutions in the required interval
4. Remember: trig functions are periodic!

****Solve for x in $[0, 2\pi)$:****

51. $\sin(x) = \frac{1}{2}$

52. $\cos(x) = -\frac{\sqrt{2}}{2}$

53. $\tan(x) = 1$

54. $\sin(x) = -\frac{\sqrt{3}}{2}$

55. $2 \cos(x) - 1 = 0$

56. $2 \sin(x) + \sqrt{3} = 0$ **Solve (may require factoring or identities):**

57. $2 \sin^2(x) - 1 = 0$

58. $\cos^2(x) - \cos(x) = 0$

59. $2 \sin^2(x) - \sin(x) - 1 = 0$
60. $\tan^2(x) - 3 = 0$
61. $\sin(2x) = \frac{\sqrt{3}}{2}$ in $[0, 2\pi)$ *Hint: Let $u = 2x$, solve for u in $[0, 4\pi)$, then find x .*
62. $\cos(2x) = -\frac{1}{2}$ in $[0, 2\pi)$ **Use identities to solve:**
63. $\sin^2(x) - \cos^2(x) = 0$ *Hint: This is $-\cos(2x) = 0$ *
64. $\sin(x) \cos(x) = \frac{1}{4}$ *Hint: Use $\sin(2x) = 2 \sin(x) \cos(x)$ *
65. $1 - \sin^2(x) = \frac{1}{2}$ **Science Connection: Resonance**
66. A system oscillates at frequency ω . Resonance occurs when the driving frequency matches the natural frequency, i.e., when: $\sin(\omega_d t) = \sin(\omega_n t)$ If $\omega_d = 2\pi$ rad/s and $\omega_n = 2\pi$ rad/s, at what times t in $[0, 1)$ seconds do the phases align (equal to each other)? **Science Connection: Standing Waves**
67. A standing wave on a string is described by: $y(x, t) = 2A \sin(kx) \cos(\omega t)$ The nodes (points that don't move) occur where $\sin(kx) = 0$. If $k = \frac{\pi}{0.5} = 2\pi \text{ m}^{-1}$, find all nodes in the interval $x \in [0, 2]$ meters. **Science Connection: Diffraction Minima**
68. In single-slit diffraction, dark fringes occur when: $\sin \theta = \frac{m\lambda}{a}$ where a is slit width, λ is wavelength, and $m = \pm 1, \pm 2, \dots$ For $a = 1000 \text{ nm}$ and $\lambda = 500 \text{ nm}$, find all angles θ where dark fringes occur (within the physical range where $|\sin \theta| \leq 1$). **General Solutions:**
69. Give the general solution (all solutions) for $\sin(x) = \frac{1}{2}$.
70. Give the general solution for $\cos(x) = 0$.